



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Main)

AIOT

28-05-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE

Time : 3 Hours

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2022

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SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

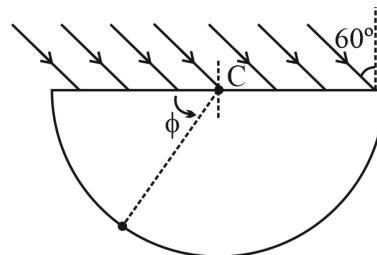
Negative Marks : -1 If wrong option is selected.

1. A particle of mass $2m$ initially at rest and placed on smooth ground explodes into two equal fragments. After explosion, they move under gravity. If after explosion difference between maximum heights of them (measured from ground) be H and difference between time taken to reach maximum height be T then impulse imparted from ground during explosion can be –

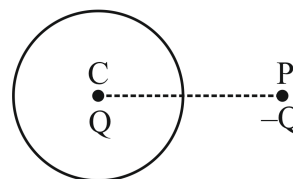
- (A) $\frac{mH}{T}$
 (B) $\frac{mH}{2T}$
 (C) $\frac{2mH}{T}$
 (D) None of these

2. Parallel light rays are falling on the plane surface of semi cylindrical rod made of glass of refractive index 2.0 at an angle of incidence of 60° in such a plane which is perpendicular to the axis of semi-cylinder as shown in figure. If ϕ is angular position where the rays are emerging out of cylindrical surface after refraction from curved surface then ϕ lies between:

[Given $\sin^{-1} \frac{\sqrt{3}}{4} = 25^\circ$]



- (A) $85^\circ < \phi < 145^\circ$
 (B) $145^\circ < \phi < 180^\circ$
 (C) $75^\circ < \phi < 105^\circ$
 (D) $25^\circ < \phi < 90^\circ$
3. A point charge Q is placed at center C of a spherical surface of radius R . Another point charge $-Q$ is placed at point P which is at distance $\frac{4R}{3}$ from center, then flux of electric field through right hemispherical surface is–



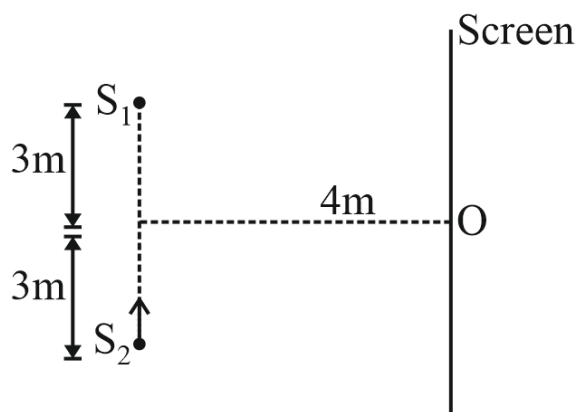
- (A) $\frac{Q}{2\epsilon_0}$
 (B) $\frac{2Q}{5\epsilon_0}$
 (C) $\frac{3Q}{5\epsilon_0}$
 (D) $\frac{Q}{5\epsilon_0}$

4. In a series RLC tuning circuit, when knob of radio is rotated, inductance of inductor varies according to $L = [0.2 + (0.1) \sin \theta] \mu\text{H}$. At what θ , circuit will be set for 100 MHz radio FM station.

(Take $\pi^2 = 10$, $C = \frac{100}{\pi^2} \text{pF}$)

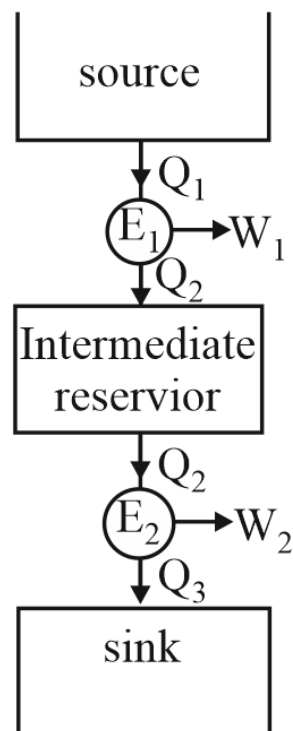
- (A) 30°
(B) 45°
(C) 60°
(D) 90°

5. S_1 and S_2 are two coherent sources of sound with zero initial phase difference. At $t = 0$, S_2 starts moving towards S_1 at speed 1 m/s as shown in figure. After what time minimum intensity will be detected at point O for the first time? [Given wavelength of sound wave = 2m]



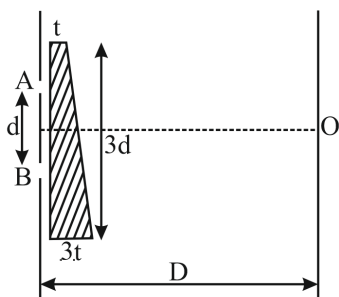
- (A) 1 sec
(B) 3 sec
(C) 2 sec
(D) 4 sec

6. Suppose two heat engines are combined in series such that the heat exhaust of the first engine E_1 is used as the heat input of the second engine E_2 as shown in figure. The efficiency of engine E_1 and E_2 are $\frac{1}{5}$ & $\frac{2}{5}$ respectively then net efficiency of combination will be.



- (A) $\frac{12}{25}$
(B) $\frac{13}{25}$
(C) $\frac{3}{5}$
(D) $\frac{1}{5}$

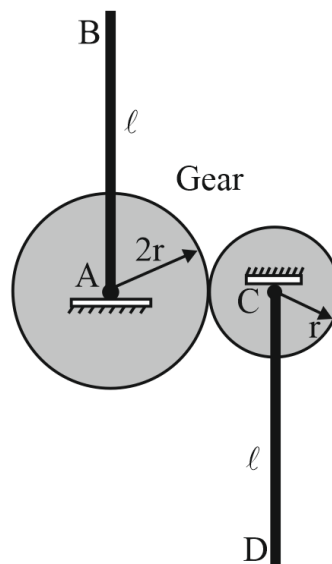
7. In a standard YDSE two slits A and B are covered by a wedge shaped thin film of refractive index μ as shown in the figure. The thickness of film varies linearly from t to $3t$ from one end to the other as shown in the figure. The distance between the two slits is d . The screen is at distance D from slits. White light is used in experiment and consider $d \ll D$. Find the distance of central maxima from the point O [centre of screen].



- (A) $\frac{2t(\mu - 1)D}{3d}$ below point O
 (B) $\frac{2t(\mu - 1)D}{3d}$ above point O
 (C) $\frac{t(\mu - 1)D}{3d}$ below point O
 (D) $\frac{2t(\mu - 1)D}{d}$ below point O
8. In a spherical symmetric charge distribution the centre of symmetry of charge distribution is at point O. Electric flux linked with a spherical Gaussian surface of radius 'r' and centered at O is given by $\phi = \phi_0 r^4$. Find the electrostatic potential energy stored within the spherical region of radius r centred at O.

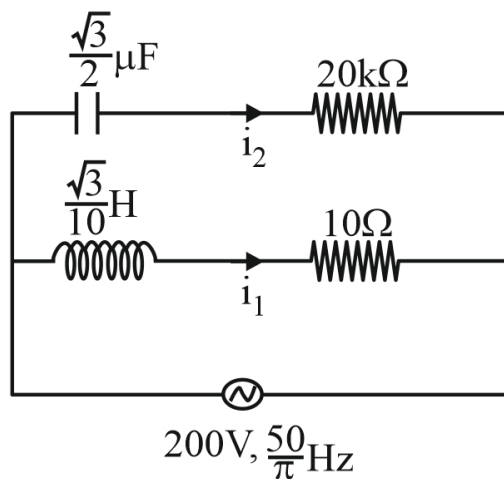
- (A) $\frac{\epsilon_0 \phi_0^2 r^7}{48\pi}$ (B) $\frac{\epsilon_0 \phi_0^2 r^7}{56\pi}$
 (C) $\frac{3\epsilon_0 \phi_0^2 r^7}{48\pi}$ (D) $\frac{\epsilon_0 \phi_0^2 r^7}{32\pi}$

9. Two uniform rods AB and CD, each of length ℓ and mass m are fixed to two gears (Which are disc's of radius $2r$ and r pivoted to fixed points A and C respectively at same horizontal level) as shown in figure. The system is in vertical plane. When rod AB is slightly rotated clockwise by a small angle, rod CD rotates in the opposite sense (anticlockwise) by another small angle such that there is no slipping and no dissipation of energy at the point of contact of the gears due to their teeth in contact. Neglecting the mass of gears, find the time period of small oscillations of the system. [g is acceleration due to gravity] [Consider $\cos \theta = 1 - \frac{\theta^2}{2!}$ neglecting higher terms]



- (A) $2\pi\sqrt{\frac{5\ell}{9g}}$ (B) $2\pi\sqrt{\frac{10\ell}{9g}}$
 (C) $2\pi\sqrt{\frac{5\ell}{g}}$ (D) $2\pi\sqrt{\frac{\ell}{g}}$

10. Find the phase difference between current i_1 and i_2 .



- (A) $\frac{\pi}{6}$
 (B) π
 (C) $\frac{\pi}{4}$
 (D) $\frac{\pi}{2}$
11. The equation of a one-dimensional transverse wave is given by $\psi = (10^{-3}) \sin \pi(10t - 3x - 2y + z)$ where x , y and z are expressed in meter and t in second. Choose the correct option.

- (A) Unit vector along the wave propagation is

$$\frac{3\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{14}}$$

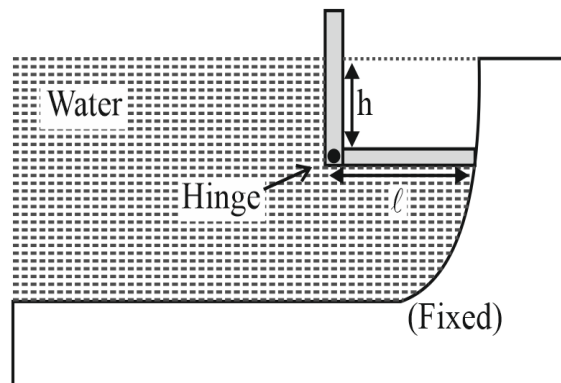
- (B) Unit vector along the wave propagation is

$$\frac{-3\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{14}}$$

- (C) Wavelength of the wave is $\sqrt{\frac{5}{7}}$ m

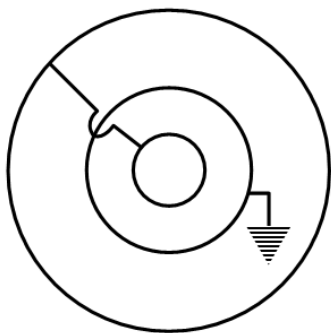
- (D) Wavelength of the wave is $\sqrt{\frac{3}{7}}$ m

12. Figure shows an L shaped hydraulic gate of uniform width, which can only rotate clockwise. The gate will start to open (rotate clockwise) if $\frac{h}{\ell} > x$, where h is the height of liquid surface above the hinge. Find value of x . [Neglect atmospheric pressure]



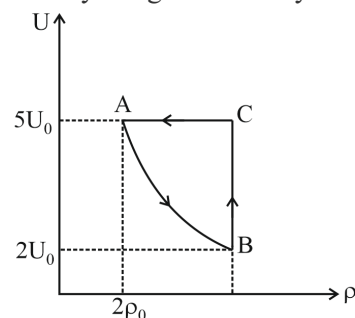
- (A) $\sqrt{3}$
 (B) $\sqrt{2}$
 (C) 1
 (D) $\sqrt{3/2}$
13. Even the most energetic of all electrons ejected from a surface by incident light of wavelength 400 nm can be just stopped before travelling 1.5 m in the direction of uniform electric field of 0.9 N/C. The work function of the surface is: [Take $hc = 12400 \text{ eV-Å}$]
- (A) 1 eV
 (B) 0.75 eV
 (C) 1.25 eV
 (D) 1.75 eV

14. There are three concentric spherical shells with radius 0.4m, 0.6m and 1.2m. Initially no charge is present on the shells. Now charge of $3\mu\text{C}$ is given to outermost shell. After this inner and outer-most shells are connected by using a thin conducting wire which passes through a small insulated hole on middle shell. The middle shell is grounded. Choose the correct option from the following:



- (A) Charge on inner surface of middle shell is $1\mu\text{C}$
 (B) Charge on inner surface of middle shell is $-2\mu\text{C}$
 (C) Charge on outer surface of outer shell is $1\mu\text{C}$
 (D) Charge on inner surface of outer shell is $2\mu\text{C}$
15. A projectile is projected from the horizontal ground from a point whose position vector is $2\hat{i} + 3\hat{j}$ metre with initial velocity $\hat{i} + 2\hat{j} + 10\hat{k}$ m/s then find the coordinates of the point where the projectile hits the ground. Consider xy plane on horizontal ground and z-axis vertically upward. ($g = 10 \text{ m/s}^2$ directed towards negative z-axis).
- (A) (7, 4, 0)
 (B) (4, 7, 0)
 (C) (5, 6, 0)
 (D) (8, 6, 0)

16. In a uniform magnetic field of $\vec{B} = 3\hat{i} + 2\hat{j} + 6\hat{k}$ Tesla, a charged particle of charge to mass ratio $2 \times 10^2 \text{ C/Kg}$ enters with velocity $\vec{V} = 4\hat{i} + 2\hat{j} - \hat{k}$ m/s. Choose the correct option. (Neglect gravity)
- (A) Frequency of revolution of the particle is $\frac{5}{\pi} \times 10^2$ cycles/s
 (B) Frequency of revolution of the particle is $\frac{3}{\pi} \times 10^2$ cycles/s
 (C) Pitch of the helical path of particle is $\frac{5\pi}{49}$ cm
 (D) Pitch of the helical path of particle is $\frac{10\pi}{49}$ cm
17. A one mole ideal monoatomic gas follows a cyclic thermodynamic process ABCA. The variation of the internal energy U with density ρ is shown in diagram. The process AB is a rectangular hyperbola. Find the net amount of heat absorbed by the gas for the cyclic process.

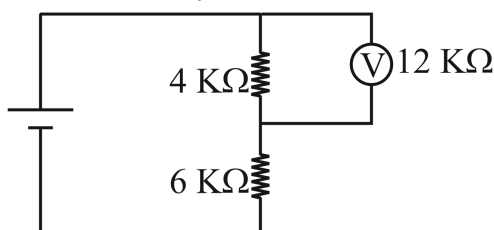


- (A) $\left[\frac{10}{3} \ln(5) - 2 \right] U_0$
 (B) $\left[\frac{5}{3} \ln(2.5) - 2 \right] U_0$
 (C) $\left[\frac{10}{3} \ln(2.5) - 2 \right] U_0$
 (D) $\left[\frac{5}{3} \ln(5) - 2 \right] U_0$

18. A rod AB of length L has uniform cross-section area S . The heat conductivity of the rod varies with the distance x measured from end A of rod as $K = K_0 \left(1 + \frac{x}{L}\right)$ where K_0 is a positive constant. If the two ends of the rod A and B are maintained at 3θ and θ temperature respectively, find the rate of heat flow in the rod.

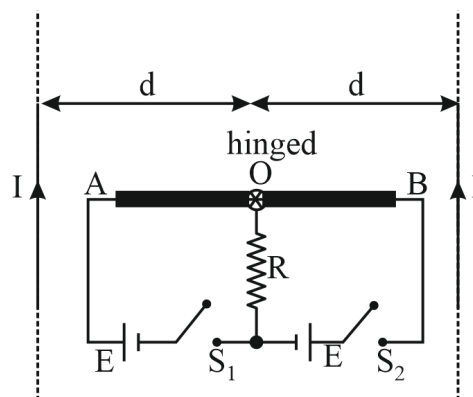


- (A) $\frac{K_0 S \theta \ln 2}{L}$
 (B) $\frac{K_0 S \theta}{L}$
 (C) $\frac{K_0 S \theta}{L \ln 2}$
 (D) $\frac{2K_0 S \theta}{L \ln 2}$
19. An ideal cell of 20V is connected to the two resistances as shown. To measure voltage across $4K\Omega$ resistance a voltmeter of $12K\Omega$ is used. The percentage error in the reading of the voltmeter will nearly be.



- (A) 19.2
 (B) 21.4
 (C) 16.7
 (D) 9.3

20. An uniform rod AB is hinged at its mid-point 'O' between two infinitely long parallel current carrying wires. The ends of the rod A & B make moving contact with connecting wires of the circuit as shown in figure. The current in both infinite wires is same and the axis of rotation of rod is perpendicular to the current carrying wires and also perpendicular to the plane of circuit. If initially switches S_1 and S_2 both are open then



- Statement 1: If only S_1 is closed, rod AB will rotate in anticlockwise
 Statement 2: If only S_1 is closed, rod AB will rotate in clockwise
 Statement 3: If only S_2 is closed, rod AB will rotate in anticlockwise
 Statement 4: If both S_1 and S_2 are closed simultaneously, rod AB remains at rest.
 Identify the correct statements

- (A) 1 and 2
 (B) 2, 3 and 4
 (C) 2 and 3
 (D) 1, 3 and 4

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

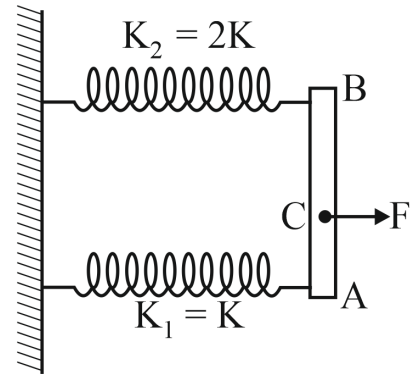
Negative Marks : -1 If wrong option is selected.

1. A fixed syringe is placed horizontally at height $h = 1\text{ m}$ from ground and a non-viscous fluid of density $\rho = 1000\text{ kg/m}^3$ is filled in syringe fitted with piston of area $A = 4\text{ cm}^2$. An orifice is made at right end of the syringe. When piston is pushed into the syringe with force $F = 10\text{ N}$, the liquid comes out of the small orifice and then following a parabolic path falls on the ground. If the speed (in m/s) with which the liquid strikes the ground is x then the x^2 is. ($g = 10\text{ m/s}^2$)

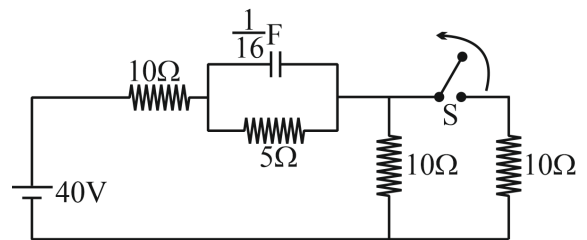


2. Height of a mountain at a place (h) is measured by measuring density of rock's material (d) forming the mountain, the elastic limit of rock's material (Y) and acceleration due to gravity (g) at that place. Percentage error in measurement of d , Y and g are respectively 2%, 1% and 2%, then find the percentage error in measurement of h .

3. A uniform rod AB of length L and mass m is attached with two springs at its two ends A and B as shown and the whole setup is placed on horizontal smooth surface. Initially the springs of spring constant K_1 and K_2 are in their natural length and the rod is horizontal. A constant force F is applied at a point C (some distance from end A) perpendicular to rod such that rod moves in horizontal plane on the smooth surface. If the rod undergoes pure translatory motion during its rightward motion then the maximum rightward displacement of the rod is $\frac{nF}{3K}$ then find n .

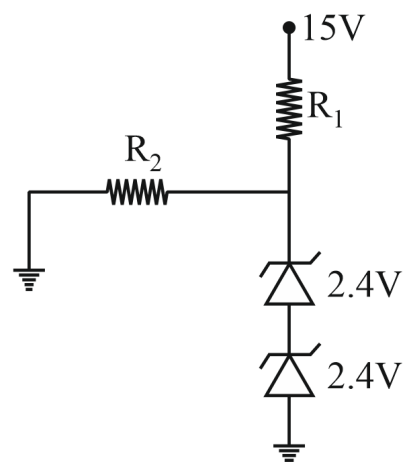


4. Initially the circuit is in steady state with switch closed. Now at $t = 0$ the switch is opened. The voltage across the capacitor $V_C(t)$ for $t > 0$ is given by $V_C(t) = A + B e^{-4t}$ Volts. Find the ratio $\frac{A}{B}$.



5. A non-uniform string of mass 50 gram and length 0.9 m has a variable linear mass density given by $\mu = kx$, where x (in meter) is the distance from one end of the string and k is a constant. Tension in the string is 16 N which is uniform. Find the time (in milli-second) required for a pulse generated at one end of the string to travel to the other end.
6. The focal length of the lens is 20 cm and 0.03 is the dispersive power of the material of a lens. Find the magnitude of difference in the focal length (in mm) of the lens for violet and red color.
7. A bulb of power 60W and having efficiency of 8.85% is considered as a point source. Find the mean of the square value (that is square of rms value) of electric field (in V/m) produced by the radiation coming from it at a distance of $\frac{10}{\sqrt{\pi}}$ metre. [Take $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$]
8. A patient is to be given a dose of medicine containing radioactive ^{18}F . The 2ml dose needs to be prepared at 9 : 00 am for it to be administered to patient exactly at 10 : 30 am. If dose administer time is changed to 11 : 15 am. Find the quantity in nearest integer of dose (in ml) that need to be prepared at 9 : 00 am. Half life of ^{18}F isotope is 90 min. (Take $\sqrt{2} = 1.41$)

9. Two identical Zener diodes of 2.4 V each are connected as shown. The resistances $R_1 = 600\Omega$ and $R_2 = 300\Omega$. Find the power dissipated in each Zener diode (in μJ).



10. In a binary-star system, two small dense stars rotate about their common center of mass, with the period of revolution 8 year for each. One star is of thrice the mass of the other and the mass of the lighter one is one-fourth the mass of the sun. The distance between the earth and the sun is R . If the distance between two stars is r , then find the ratio of r/R . [Assume the orbits to be circular]

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

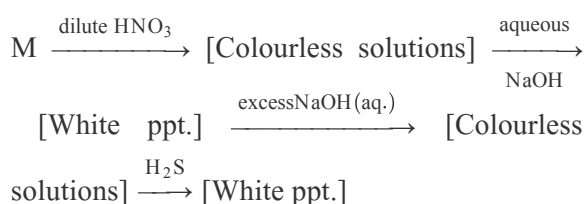
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. How long (approximate) should water be electrolysed by passing through 100 amperes current so that the oxygen released can completely burn 27.6 g of diborane?
(Atomic weight of B = 10.8 u)

- (A) 1.6 hours (B) 6.4 hours
(C) 0.8 hours (D) 3.2 hours

2. A metal M and its compound can give the following observable changes in a consequence of reactions



- (A) Mg (B) Pb (C) Zn (D) Sn

3. A complex of molecular formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ has green colour. 0.1 mole of the complex when treated with excess of AgNO_3 gave 28.7 g of white precipitate of AgCl (Molar mass = 143.5 g). The formula of the complex would be

- (A) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
 (B) $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$
 (C) $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl} \cdot 2\text{H}_2\text{O}$
 (D) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3]3\text{H}_2\text{O}$

4. The simultaneous solubility of AgCN ($K_{\text{sp}} = 2.5 \times 10^{-16}$) and AgCl ($K_{\text{sp}} = 1.6 \times 10^{-10}$) in 1 M NH_3 (aq) are respectively

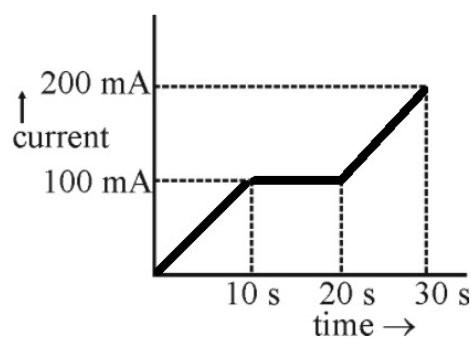
Given $K_f[\text{Ag}(\text{NH}_3)_2]^+ = 10^7$

- (A) 1.58×10^{-3} , 1.26×10^{-5}
(B) 0.04, 6.25×10^{-8}
(C) 5.78×10^{-8} , 0.037
(D) 0.037, 5.78×10^{-8}

5. Which of the following expression is correct for packing fraction of NaCl if anions from one face diagonal are removed?

$$\begin{array}{ll} \text{(A)} \quad \frac{\frac{13}{8}\pi r_-^3 + \frac{16}{3}\pi r_+^3}{8(r_+ + r_-)^3} & \text{(B)} \quad \frac{\frac{13}{3}\pi r_-^3 + \frac{16}{3}\pi r_+^3}{8(r_+ + r_-)^3} \\ \text{(C)} \quad \frac{\frac{16}{3}\pi r_-^3 + \frac{13}{3}\pi r_+^3}{8(r_+ + r_-)^3} & \text{(D)} \quad \frac{\frac{4}{3}\pi r_-^3 + \frac{13}{3}\pi r_+^3}{8(r_+ + r_-)^3} \end{array}$$

6. In a Cu-voltameter, mass deposited in 30s is 'm' gm. If the time-current graph is shown in the following figure



What is the electrochemical equivalent of Cu?

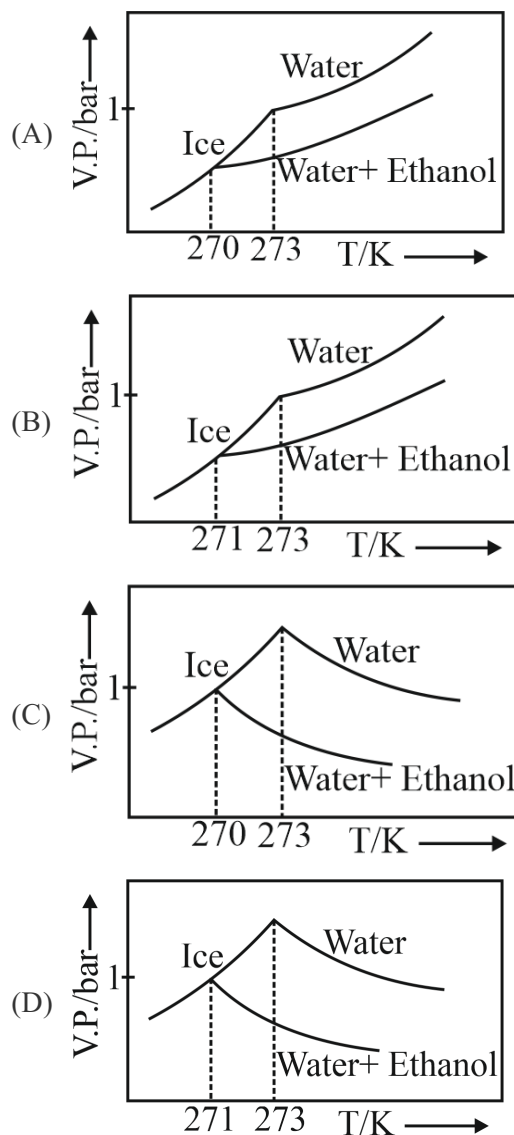
- (A) $\frac{m}{2}$ (B) $\frac{m}{3}$
(C) $\frac{m}{4}$ (D) $\frac{m}{63.5}$

7. Which of the following sets of reactants is used for preparation of paracetamol from phenol?
- (A) HNO_3 , H_2/Pd , $(\text{CH}_3\text{CO})_2\text{O}$
 (B) H_2SO_4 , H_2/Pd , $(\text{CH}_3\text{CO})_2\text{O}$
 (C) $\text{C}_6\text{H}_5\text{N}_2\text{Cl}$, SnCl_2/HCl , $(\text{CH}_3\text{CO})_2\text{O}$
 (D) $\text{Br}_2/\text{H}_2\text{O}$, Zn/HCl , $(\text{CH}_3\text{CO})_2\text{O}$
8. The minimum voltage required to electrolyse alumina in the Hall-Heroult process is
 Given, $\Delta G_f^\circ (\text{Al}_2\text{O}_3) = -1520 \text{ kJ mol}^{-1}$
 $\Delta G_f^\circ (\text{CO}_2) = -394 \text{ kJ mol}^{-1}$
- (A) 1.575 V (B) 1.60 V
 (C) 1.312 V (D) -2.62 V
9. Find the final product 'E'
- $\text{CH}_2\text{OH} - \text{CH} = \text{CH}_2 \xrightarrow[\text{(ii) NaSO}_3\text{H}]{\text{(i) OsO}_4} \text{(A)} \xrightarrow{\text{KHSO}_4}$
 $\text{(B)} \xrightarrow{\text{N}_2\text{H}_4/\text{OH}^-} \text{(C)} \xrightarrow[\text{CCl}_4]{\text{NBS}/h\nu} \text{(D)} \xrightarrow{(\text{CH}_3)_2\text{CuLi}} \text{(E)}$
- (A) But-2-ene (B) But-1-ene
 (C) Isopentene (D) Pent-2-ene
10. Match List-I with List-II and select the correct answer.

	List-I		List-II
P.	α -D and β -D glucose	1.	Enantiomers
Q.	D-glucose and D-galactose	2.	Anomers
R.	Erythrose and threose	3.	Epimers
S.	D-glyceraldehyde and L- glyceraldehyde	4.	Diastereomers

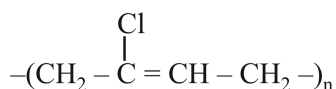
- (A) P-3, Q-2, R-1, S-4 (B) P-2, Q-3, R-4, S-1
 (C) P-3, Q-2, R-4, S-1 (D) P-2, Q-3, R-1, S-4

11. Pure water freezes at 273 K and 1 bar. The addition of 34.5 g of ethanol to 500 g of water changes the freezing point of the solution. Use the freezing point depression constant of water as 2 K kg mol^{-1} . The figures shown below represent plots of vapour pressure (V.P.) versus temperature (T).
- Among the following, the option representing change in the freezing point is (assume ethanol to be non-volatile)



12. Which of the following is not correctly matched?

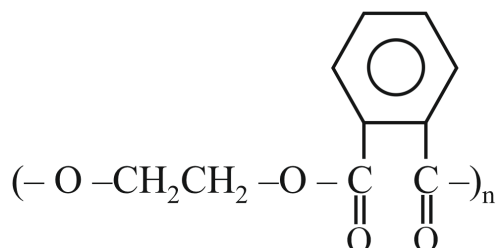
(A) Neoprene :



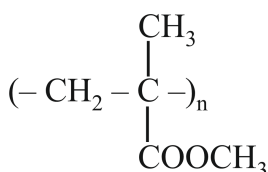
(B) Nylon-66:



(C) Terylene :



(D) Polymethyl methacrylate (PMMA):



13. Which of the following statement is true?

(A) SbF_4^- and SF_4 are isostructural

(B) In IOF_5 the hybridization of central atom is sp^3d^2

(C) Double bond(s) in SOF_4 and XeO_3F_2 , is/are occupying equatorial position(s) of their

(D) All of these

14. Match List-I with List-II and select this correct answer using the codes given below the lists.

	List-I (Type of glass)		List-II (Property/use)
(p)	Borosilicate glass	1.	Very high transparency
(q)	Calcium-alkali silicate glass	2.	Cheap laboratory glass wares
(r)	Lead glass	3.	Optical glass
(s)	Soda glass	4.	Domestic glass for windows
		5.	Low coefficient of expansion

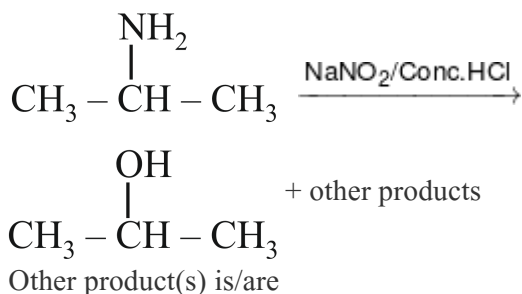
(A) (p)-2, (q)-5, (r)-3, (s)-4

(B) (p)-2, (q)-1, (r)-5, (s)-4

(C) (p)-5, (q)-2, (r)-1, (s)-3

(D) (p)-5, (q)-4, (r)-3, (s)-2

15. Consider the following reaction

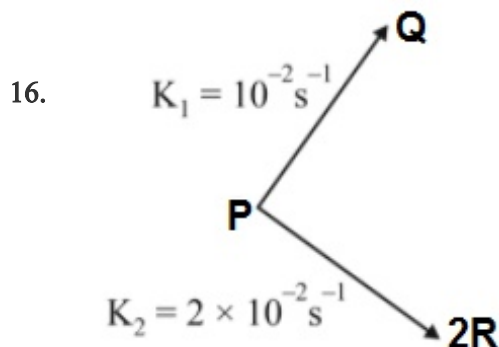


(A) 2-chloro propane

(B) Propene

(C) Isopropyl nitrite

(D) All of these



Let the initial concentration of P is 1 M, the concentration of P after 33.33 sec is approximate equal to

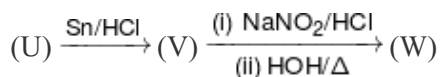
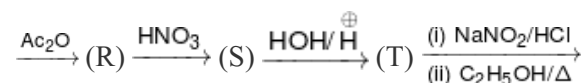
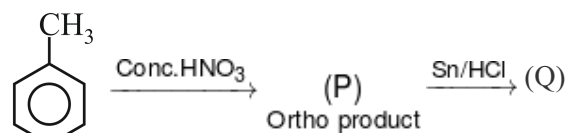
- (A) $\frac{1}{e}$ M
(B) $\frac{2}{e}$ M
(C) $\frac{1}{e^2}$ M
(D) e M

17. An electron practically at rest, is initially accelerated through a potential difference of 100 volts. It then has a de Broglie wavelength = $\lambda_1 \text{ \AA}$. It then get retarded through 19 volts and then has a wavelength $\lambda_2 \text{ \AA}$. It further retardation through 32 volts changes the wavelength to $\lambda_3 \text{ \AA}$. What is the value of $\frac{\lambda_3 - \lambda_2}{\lambda_1}$?
- (A) $\frac{20}{41}$
(B) $\frac{10}{63}$
(C) $\frac{20}{63}$
(D) $\frac{10}{41}$

18. The correct order of boiling point is

- (A) $T_2 < D_2 < H_2$
(B) n-pentane < neo-pentane
(C) Xe < Ar < He
(D) m-nitrophenol > o-nitrophenol

19. Identify the final product 'W' in the given reaction sequence



- (A) Ortho-methyl phenol
(B) Meta-methyl phenol
(C) Toluene
(D) Meta-nitro toluene

20. Consider the following metallurgical processes
- (I) Heating impure metal with CO and distilling the resulting volatile carbonyl (boiling point. 43°C) and finally decomposition at $150^{\circ} - 200^{\circ}\text{C}$ to get the pure metal
- (II) Heating the sulphide ore in air until a part is converted to oxide and then further heating in the absence of air to let the oxide react with unchanged metal sulphide
- (III) Electrolysis of the molten electrolyte containing approximately equal amounts of the metal chloride and NaCl to obtained the metal
- The processes used for obtaining magnesium, nickel and copper are respectively
- (A) (I), (II) and (III)
(B) (II), (III) and (I)
(C) (III), (I) and (II)
(D) (II), (I) and (III)

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

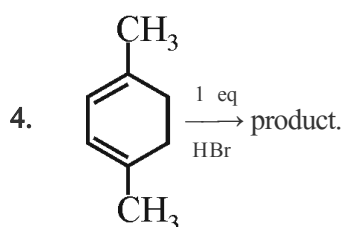
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. An unknown compound **A** $C_8H_{10}O_3$ on acetylation with CH_3COCl/Py forms acetyl derivative of **A** whose MW is 280. **A** on treat with CH_2N_2 gives methyl ether of **B** having MW 182. If the number of phenolic hydroxyls and alcoholic hydroxyls in the compound **A** are **X** and **Y** respectively. Find the sum of **X** + **Y** here?
2. A sample of H_2SO_4 (density 1.80 g mL^{-1}) is labelled as 78.4% by weight. What volume (in mL) of this acid has to be used to make 1 L of $0.9 \text{ M } H_2SO_4$?
3. The possible number of stereo isomers for the complex $[Co(EDTA)]^-$ is :-

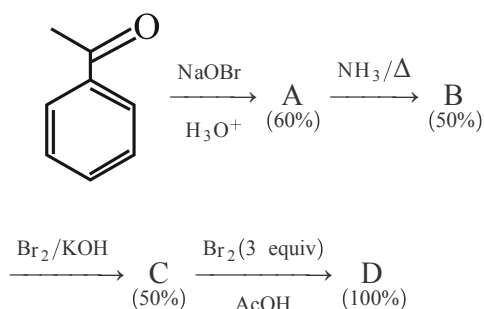


No of allylic hydrogen atoms in thermodynamically controlled product is "X"

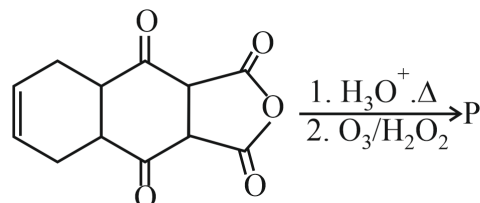
No of allylic hydrogen atoms in kinetically controlled product is "Y"

The value of $|X + Y|$ is

5. Serotonin is a famous tranquilizer if it contains 'X' number of lone pairs of electrons and 'Y' number of π -bonds. The sum of $X + Y$ is equal to
6. In the following reaction sequence, the amount of **D** (in g) formed from 10 moles of acetophenone is _____.
(Atomic weights in g mol^{-1} : $H = 1$, $C = 12$, $N = 14$, $O = 16$, $Br = 80$).
The yield (%) corresponding to the product in each step is given in the parenthesis)



7. The total number of carboxylic acid groups in the product **P** is



8. 0.28 g of solid adsorbent is dropped in a container of 1.5 L N_2 gas at 1 atm and 300 K. The pressure of N_2 is reduced to 20%. The mass of nitrogen (in g) adsorbed per gram of adsorbent is _____.
[Given: $R = 0.08 \text{ L atm K}^{-1} \text{ mol}^{-1}$]

9. How many of the given statements are correct ?
- (i) Solubility of gases in liquids decreases as temperature of solution increases
 - (ii) Disinfectants are applied to inanimate.
 - (iii) Sols of starch, gum, clay and charcoal are negative sols
 - (iv) Froth floatation is never used for the beneficiation of copper glance
 - (v) Potassium permanganate is prepared by fusion of MnO_2 with KOH and O_2 or KNO_3 . The KNO_3 in this reaction act as reducing agent.
 - (vi) Size of Pr^{+3} is more than Tb^{+3}
 - (vii) Hydroxide of lanthanoids are basic in nature
 - (viii) As temperature increases in a 1st order reaction, half life of reaction remains same
 - (ix) Frenkel defect is also called dislocation defect
 - (x) In ferrimagnetic substances, magnetic moment of all domains are aligned in same direction
 - (xi) Antiseptics are applied to the living tissues
 - (xii) Aspirin is narcotic analgesics
 - (xiii) Aspartame is artificial sweetener
10. 4.0 moles of an ideal monoatomic gas undergoes adiabatic free expansion from (10L, 300K) to 20L. The value of ΔS (in cal/K) of gas is [$\ln 2 = 0.70$]

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- The locus of the point z which moves such that $2\arg\left(\frac{z-i+3}{z+3i-1}\right) = \pi$ is -
 (A) A straight line passing through the points $(3-i)$ and $(-1+3i)$
 (B) A straight line passing through the points $(-3+i)$ and $(1-3i)$
 (C) A semi-circle passing through the points $(-3-i)$ and $(1-3i)$
 (D) A part of circle with centre at the point $(-1-i)$ and radius $2\sqrt{2}$
- If the exact value of $E = \left(\int_0^{\pi/2} (\cos x)^{2022} dx\right) \left(\int_0^{\pi/2} (\cos x)^{2023} dx\right)$ is $\frac{\pi}{k}$ then k is equal to
 (A) 4046
 (B) 2023
 (C) 2022
 (D) 4044

- $\lim_{n \rightarrow \infty} \frac{(1^{2022} + 2^{2022} + 3^{2022} + \dots + n^{2022})n}{1^{2023} + 2^{2023} + \dots + n^{2023}}$ is equal to $\frac{a}{b}$ where a and b are coprime then $a+b =$
 (A) 4045 (B) 4046
 (C) 4047 (D) None of these
- If α, β, γ are the roots of $x^3 - 2x^2 + 4x - 6 = 0$ then find the value of $\left(\alpha\beta - \frac{2}{\gamma}\right) \left(\beta\gamma - \frac{2}{\alpha}\right) \left(\gamma\alpha - \frac{2}{\beta}\right)$
 (A) $\frac{32}{3}$
 (B) $\frac{64}{3}$
 (C) $\frac{16}{3}$
 (D) $\frac{10}{3}$
- Number of words which can be formed by using all the letters of the word 'ALLENBBSR' such that the word 'BBSR' is always there and occupies among the first six positions only, is
 (A) 120 (B) 160
 (C) 180 (D) 210
- Sum of the series $S = 1 + 2\left(\frac{10}{11}\right) + 3\left(\frac{10}{11}\right)^2 + \dots$ upto infinite terms, is equal to
 (A) 121
 (B) 120
 (C) 111
 (D) 110

7. The false statement among the following is -

- (A) $p \wedge (\sim p)$ is a contradiction
 (B) $(p \rightarrow q) \Leftrightarrow (\sim q \rightarrow \sim p)$ is a contradiction
 (C) $\sim(\sim p) \Leftrightarrow p$ is a tautology
 (D) $p \vee (\sim p)$ is a tautology

8. The mean of n items is $\bar{x}$. If each item is successively increased by $3, 3^2, 3^3, \dots, 3^n$, then new mean equals -

- (A) $\bar{x} + \frac{3^{n+1}}{n}$
 (B) $\bar{x} + 3 \left(\frac{3^n - 1}{2n} \right)$
 (C) $\bar{x} + \frac{3^n}{n}$
 (D) $\bar{x} + 3 \cdot \left(\frac{3^{n-1} - 1}{2n} \right)$

9. If solution of the differential equation $x \left(\frac{d^3 y}{dx^3} \right) + \left(\frac{d^2 y}{dx^2} \right) + 2x \left(\frac{dy}{dx} \right) - y + 1 = 0$ is a straight line of the form $y = mx + c$, then number of such possible lines is

- (A) 1 (B) 2
 (C) 3 (D) Infinite

10. The range of ' α ' for which all the points of local extrema of the function $f(x) = x^3 - 3\alpha x^2 + 3(\alpha^2 - 1)x + 1$ lie in the interval $(-2, 4)$, is

- (A) $(-1, 3)$
 (B) $(3, 4)$
 (C) $(-4, -2)$
 (D) $(-2, -1)$

11. General solution of differential equation $e^x \frac{dy}{dx} + e^x y = 1$, is

- (A) $y = e^x + ce^{-x}$
 (B) $y = xe^x + ce^x$
 (C) $y = e^{-x} + ce^x$
 (D) $y = xe^{-x} + ce^{-x}$

12. If A and B are non zero matrices of order 3 such that $3A + 2B = A^T$, then $\det(A + B)$ is

- (A) 1 (B) -1
 (C) 2 (D) 0

13. Sum of the series $\sum_{r=0}^{10} (-1)^r {}^{10}C_r \left(\frac{1}{3^r} + \frac{8^r}{3^{2r}} \right)$ is

- (A) $\frac{2^{10} + 1}{3^{20}}$
 (B) $\left(\frac{2}{3} \right)^{10} - \left(\frac{1}{9} \right)^{10}$
 (C) $\frac{3^{10} - 1}{3^{20}}$
 (D) $\frac{6^{10} + 1}{3^{20}}$

14. The centre of family of circles cutting the family of circles

$$x^2 + y^2 + 4x \left(\lambda - \frac{3}{2} \right) + 3y \left(\lambda - \frac{4}{3} \right) - 6(\lambda + 2) = 0$$

orthogonally, lies on

- (A) $x - y - 1 = 0$
 (B) $4x + 3y - 6 = 0$
 (C) $4x + 3y + 7 = 0$
 (D) $3x - 4y - 1 = 0$

15. The greatest integral value of 'a' for which the equation $(x^2 + x)^2 + a(x^2 + x) + 4 = 0$ has 4 distinct real roots.

(A) -4 (B) -3 (C) 3 (D) -5

16. The ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ has a double contact with a circle at the extremity of latus rectum. The point of contact lying in first and fourth quadrant. Then

(A) Centre of circle is (0, 0)
 (B) Centre of circle is $\left(\frac{1}{4}, 0\right)$
 (C) Radius of circle is $3\sqrt{5}$
 (D) Radius of circle is $\frac{3\sqrt{5}}{2}$

17. The number of ways in which one or more balls can be selected out of 5 white, 6 green and 3 blue balls are :

(Balls of same colour are identical)

(A) 168
 (B) 167
 (C) 169
 (D) 166

18. The value of $\sum_{r=1}^9 \sin^2 \frac{r\pi}{10}$ is :

(A) 1
 (B) $\frac{5}{2}$
 (C) 5
 (D) $\frac{9}{2}$

19. If $\hat{a}$, $\hat{b}$ and $\hat{c}$ are unit vectors such that $\hat{a} \wedge \hat{b} = \hat{b} \wedge \hat{c} = \hat{c} \wedge \hat{a} = 60^\circ$ then

$$(\hat{a} \times \hat{b}) \times (\hat{a} + \hat{b} + \hat{c}) = \lambda (\hat{b} - \hat{a}), \text{ where } \lambda \text{ is}$$

(A) -2
 (B) 2
 (C) 1
 (D) -1

20. The value of $\int_0^{\pi/2} \frac{(\cos 2\theta + 1)^2}{\cos^4 \theta + \sin^2 2\theta} d\theta$ is

(A) $\frac{\pi}{6}$ (B) $\frac{2\pi}{3}$
 (C) $\frac{3\pi}{2}$ (D) None of these

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Let $f(x) = (x^3 + (a-1)x^2 + (b-a)x - b) |x^2 + 8x + 6|$ where $a, b \in \mathbb{R}$. If $f(x)$ is derivable in $(-\infty, \infty)$ then $(a+b)$ is equal to

2. If $g(x) = (\sin x) \int_0^x \cos t \, dt + \cos^2 x - \frac{x^2}{2} + \int_0^x t \, dt + 4$
and area bounded by the curve $y = 3g(x) (x^2 - 3x + 2)$
and the x-axis is A, then $10A$ is
3. A box contains 12 mangoes out of which 5 are rotten and rest are good. Two mangoes are randomly taken out together. If it is known that atleast one of them is good, then the probability that both are good is (p/q) , where p, q are coprime numbers. The value of $|p - q|$ is :
4. Let AB is a normal chord of parabola $y^2 = 4x$,
with foot of normal as $A(1, -2)$.
If $AB = a\sqrt{b}$, where a is a natural number and b is prime number, then $\frac{a}{b}$ is equal to
5. The integer 'n' for which the value of $\lim_{x \rightarrow 0} \frac{(1 - \cos x)(e^x - \cos x) - \frac{x^3}{2}}{x^n}$ is a finite non zero number, is
6. If $f(x) = \sin^{-1}x$ and $g(x) = [\cos(\sin x)]$, where $[\cdot]$ denotes greatest integer function. Then number of element(s) in the range of $f(g(x))$ is
7. If normal at point $(4, 1)$ of the curve $xy = 4$ intersects the curve again at the point (α, β) , then β/α is equal to
8. The value of
$$\frac{(1 + \sqrt{3}\tan 1^\circ)(1 + \sqrt{3}\tan 2^\circ)(\tan 1^\circ + \tan 59^\circ)(\tan 2^\circ + \tan 58^\circ)}{(1 + \tan^2 1^\circ)(1 + \tan^2 2^\circ)}$$
 is -
9. If $\sum_{r=1}^{19} \frac{3(r+1)r+1}{(r+1)(r^5+2r^4+r^3)} = \frac{a}{b}$ (a & b are co-prime numbers), then $(b - a)$ is
10. Let $\int_0^9 e^x (1 + \{x\} + \{x\}^2) dx = I$ and $\int_0^1 e^x (1 + \{x\} + \{x\}^2) dx = J$, if $\frac{I}{J} = \sum_{n=0}^k e^n$, then k is equal to (where $\{.\}$ denotes fractional part function)

Space for Rough Work

